Human Coder Guide

Digital Task Intensity (DTI) of Indian Occupations — validation of occupation scores
Version 1.0 · October 2026

Research team	Dr. Anuja Nair (Institute of Technology), Dr. Gunjan Anand and Dr. Samik Shome (Institute of Management), Nirma University, Ahmedabad
Your coder ID	_____
Coordinator / queries	_____ (name, e-mail, phone)
Practice set due	_____
Main set due	_____

1. What we are asking you to do

Thank you for agreeing to be a coder. You will read short official descriptions of Indian occupations from the **National Classification of Occupations (NCO-2015)**. For each one you will rate how much digital work the job involves, on five dimensions.

Your ratings are the human benchmark for our study. We will compare them with scores produced by a computer-based method, so it matters that you **follow the rules in this guide exactly** and work **independently**.

> **In short:** read the occupation text → give five scores from 0 to 4 → copy the words that support each score → answer a few short follow-up questions. About **3–5 minutes per occupation**.

2. How the work is organised

Stage	What happens	Size
1. Read this guide	Read the whole guide, including the worked example in Section 7.	About 20 minutes
2. Practice (calibration) set	Code a small practice set. The team then holds a short meeting to discuss differences and clarify the rules. Practice scores are **not** used in the results.	About 30 occupations
3. Main set	Code the main set on your own. **No discussion with other coders** until everyone has submitted.	_____ occupations
4. Feedback form	Fill the short feedback form (Section 9) once you finish.	5 minutes

You will receive a **coding sheet (Excel)**, one row per occupation, with the code, title and description already filled in. Enter your answers in that sheet. A printable answer form is at the end of this guide, in case you prefer paper for the practice set.

3. Coding rules

1. **Score only from the text you are given**: title, description and tasks. Do not score from what you think the job should involve or from what a particular employer does. (Question 11 is the only place for your own view.)

2. **If the text says nothing about a dimension, score 0.** Do not guess. Mark Question 1 as "Partly" or "No" if the text is thin.
3. **Score the typical worker** in the occupation, not the most advanced or most modern one.
4. **Score each dimension separately.** A high score on one dimension does not mean high scores on the others.
5. **Plain voice phone calls do not count as digital.** Messaging, e-mail, apps and video calls do.
6. **Use the anchors.** When unsure between two scores, compare with the anchor examples and pick the closer one. Note the doubt in Question 12.
7. **Work independently.** Do not discuss items with other coders, do not search for how others scored, and do not use AI tools (ChatGPT etc.) to help.
8. **Keep the material confidential.** Do not share the coding sheet or your scores outside the research team.
9. **Take breaks.** Rating quality falls after long sessions; we suggest no more than about 40 occupations at a sitting.

4. The 0–4 scale (same for every dimension)

Score	Label	Meaning
0	None	Not mentioned and not needed to do the job.
1	Incidental	Occasional, simple use. The job can be done without it.
2	Regular, basic	Used routinely as part of the job, with standard tools.
3	Regular, advanced	Frequent use of complex or job-specific digital tools.
4	Core / specialist	This digital activity *is* the main work of the occupation.

5. The five dimensions

The examples are guides, not a complete list. Score the occupation in front of you against the meaning of each level.

D1. Basic digital operation

Routine use of computers, mobile phones, tablets, POS or billing machines, and standard software (word processing, spreadsheets, data-entry screens).

0	No device use described (e.g. manual field or construction labour).
1	Occasional simple use (e.g. shop salesperson who sometimes uses a billing machine).
2	Daily use of standard tools (e.g. office clerk entering records).
3	Heavy use of several standard applications through most of the day (e.g. bank teller working in core banking software).
4	Operating the computer is the job (e.g. computer or data-entry operator).

D2. Digital information processing

Searching, recording, managing, interpreting or analysing information in digital form.

0	No digital information handled.
1	Occasionally looks up information online or on a system.
2	Keeps digital records or registers (e.g. stock records in accounting software).
3	Prepares reports or analysis from digital data or databases (e.g. accountant).
4	Analysing digital data is the main work (e.g. statistician, data analyst).

D3. Digital communication and collaboration

E-mail, messaging, video calls, and working through digital platforms: CRM, ticketing, e-commerce or gig apps, government portals (e.g. GeM).

0	No digital communication (face-to-face or voice calls only).
1	Occasional messages (e.g. WhatsApp or SMS with a supervisor).
2	Regular e-mail or messaging with clients or colleagues.
3	Work is organised through a platform (e.g. CRM, delivery or booking app, procurement portal).
4	The role is digital interaction (e.g. online customer support, social media manager).

D4. Advanced or specialised digital tools

Occupation-specific software, programming, CAD, GIS, analytics tools, and computer-controlled machines or systems (CNC, PLC, medical imaging, automated plant controls).

0	No specialised digital tools.
1	Operates equipment with simple digital controls (e.g. a machine with a digital display and preset buttons).
2	Routinely uses one specialised software or runs preset programs (e.g. hospital records system, preset CNC program).
3	Sets up, configures or programs specialised tools (e.g. CAD drafting, CNC programming, GIS mapping).
4	Builds or maintains such systems (e.g. software developer, systems or network engineer).

D5. Digital problem solving and content creation

Using digital technology to create content (documents, designs, video, code, models) or to solve non-routine problems.

0	No digital content or problem solving.
1	Makes simple digital documents (e.g. typing a letter or a basic list).
2	Produces routine digital content from templates (reports, slides, posters).
3	Creates original digital content or solves non-routine problems with digital tools (e.g. graphic designer, IT support technician).
4	Creating digital products or solutions is the core work (e.g. software architect, animator, data scientist).

6. The questions you answer for each occupation

Q	Question	How to answer
Q1	Is the description detailed enough to judge digital tasks?	Yes / Partly / No
Q2	D1 Basic digital operation	Score 0–4
Q3	D2 Digital information processing	Score 0–4
Q4	D3 Digital communication and collaboration	Score 0–4
Q5	D4 Advanced or specialised digital tools	Score 0–4
Q6	D5 Digital problem solving and content creation	Score 0–4
Q7	**Evidence.** For each dimension you scored 1 or more, which words in the text support your score?	Copy the exact words from the text (a few words is enough)
Q8	**Confidence** in each of your five scores	1 = low, 2 = medium, 3 = high
Q9	**Overall**: without adding up your answers, how digital is this occupation overall?	0 (not at all) to 10 (entirely digital)
Q10	Is digital work **core** to this job, or a **supporting** tool?	Core / Supporting / Not relevant
Q11	**Your own view:** in India today, is this occupation more digital than the text suggests, about the same, or less?	More / Same / Less *(the only question where you may use your own knowledge)*
Q12	Was anything unclear, or did the case not fit the rules?	Short comment (optional)

7. Worked example

Accounts Clerk illustrative text, not an actual NCO entry

"Records financial transactions in ledgers and accounting software; prepares vouchers, invoices and bank reconciliation statements; maintains digital files of bills and receipts; corresponds with vendors by e-mail regarding payments; prints and files monthly summary reports."

Question	Answer	Evidence / reason
Q1 Enough detail?	Yes	Tasks are listed clearly.
Q2 D1 Basic operation	2	"accounting software", "digital files": daily standard use, not the whole job.
Q3 D2 Information processing	3	"bank reconciliation statements", "monthly summary reports": prepares reports from digital records.
Q4 D3 Communication	2	"corresponds with vendors by e-mail": regular e-mail.
Q5 D4 Specialised tools	2	"accounting software": one specialised package used routinely.
Q6 D5 Content / problem solving	1	"prepares vouchers, invoices": simple documents, no original content.
Q8 Confidence	3, 3, 3, 2, 2	Less sure on D4 and D5.
Q9 Overall	5	Moderately digital.
Q10 Core or supporting	Supporting	The job is accounting; digital tools support it.
Q11 Reality check	More	Most clerks now also use online banking and GST portals.
Q12 Comment	—	

Note: Q11 is "More" but the scores in Q2–Q6 stay text-based. Your own knowledge goes only in Q11.

8. Common mistakes to avoid

- Giving every dimension the same score because the job "feels" digital. Score each one on its own.
- Scoring above 0 when the text never mentions the activity.
- Counting voice phone calls as digital communication.
- Scoring the best-equipped worker (a large private firm) instead of the typical worker.
- Leaving Q7 (evidence) blank for a score of 1 or more.
- Changing earlier answers after seeing later occupations. If you notice a real error, correct it and note it in Q12.

9. Feedback form (fill once, after the main set)

Coder ID	
1. Which dimension was hardest to score, and why?	
2. Were any anchor examples misleading or missing?	
3. Did any two dimensions feel like they overlapped? Which?	
4. Roughly how many minutes did one occupation take?	
5. Any other comments on the rules or the material?	

10. Coder profile (fill once, before starting)

Coder ID	
Field	HR · Economics · Computer science · Other: _____
Years of experience with Indian occupations or labour markets	
I have read this guide and completed the practice set	Yes · No
I will code independently and keep the material confidential	Signature: _____ Date: _____

Printable answer form (one occupation)

Coder ID _____ NCO-2015 code _____

Occupation title _____

Question	Answer (circle)	Confidence (1–3)
Q1 Enough detail?	Yes / Partly / No	
Q2 D1 Basic digital operation	0 1 2 3 4	
Q3 D2 Information processing	0 1 2 3 4	
Q4 D3 Communication & collaboration	0 1 2 3 4	
Q5 D4 Specialised digital tools	0 1 2 3 4	
Q6 D5 Problem solving & content	0 1 2 3 4	

Q7 Evidence (exact words from the text, for each score of 1 or more)

D1	
D2	
D3	
D4	
D5	

Q9 Overall digital intensity (0–10)	0 1 2 3 4 5 6 7 8 9 10
Q10 Digital work is…	Core / Supporting / Not relevant
Q11 In India today, compared with the text, this job is…	More digital / About the same / Less digital
Q12 Comment	